Measuring Generalization Surprise in AI

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Abstract

AI systems generalize in unpredictable ways. An AI system might simultaneously solve complex theoretical physics problems, while making errors in simple grade school math. AI evaluation suites must then calibrate users'—and policymakers'—expectations to this variability. Most evaluation efforts, however, do not account for how humans might interpret evaluations. To address this, we introduce *generalization surprise*, a metric approximating how much people would be surprised by an AI system's performance on a new problem, given previous reported evaluation results. We operationalize this notion of surprise using a model of human expectations and evaluate it over five families of AI systems on a newly introduced suite of 85 diverse benchmarks. We discover that, despite scoring comparably on standard benchmarks, the latest AI systems can differ markedly in generalization surprise. Even closed frontier systems exhibit myriad generalization failures, especially in tasks outside formal sciences. Our generalization surprise metric can help address this by scalably identifying surprising failure patterns, valuable for smoothing out generalization irregularities and enhancing evaluation construction—we accordingly introduce 'SurpriseBench,' a simple benchmark where even frontier AI systems surprisingly score at most 20%. Further, unlike conventional leaderboards, we can combine both system error and surprise into a *generalization loss* metric that resists score-hacking (e.g., training on test data) that inflate performance without improving generalization. Additionally, our framework reveals systematic gaps in existing model cards of frontier systems that limit people's ability to form reliable expectations about AI's generalization, for which we propose improvements. Our conceptual, methodological, and empirical contributions represent a major advance in AI evaluation and data-centric training toward more generalizable and safe general intelligence.

1 Introduction

Proper evaluations have the potential to empower people to better understand AI models—in turn allowing stakeholders, like users and policymakers, to better steer the trajectory of future AI systems [1–4]. Due to the high stakes associated with frontier AI systems, it has become an industry standard to accompany their releases with evaluation suites reported in model cards—each typically consisting of a set of benchmarks aggregating a wide range of challenging tasks [5–8]. Paradoxically, despite their intended role in informing users and policymakers, current evaluations often fail to characterize myriad unexpected behaviors and failure modes [9–17], forcing many to rely on ad-hoc out-of-distribution "vibe checks" to superficially intuit the capabilities of AI systems [18].

We attribute this disconnect to two unsolved, intertwined challenges. The first challenge is the *opaqueness of generalizability* of AI systems, where they can fail to maintain the reported performance on benchmark suites when used in untested tasks that are perceived as similar by humans. Achieving this human-centric notion of generalization is vital to reduce undesirable surprises since AI usage during deployments is highly dynamic and diverse, unlike existing evaluations that tend to prioritize subtasks that are most convenient to evaluate and hill-climb within given domains [19–21]. The second challenge is that existing *evaluation suites overlook the importance of reducing surprises* when releasing new frontier AI systems. For example, when an evaluation is biased toward certain domains, they provide limited information about how the system will behave in untested use cases encountered during deployment, specially for those use cases that humans perceive as unrelated to the evaluated domains.

Addressing these two challenges is critical for *patching surprising generalization failures* and ensuring that humans can *reliably form expectations about AI behavior*, fostering a broader positive real-world impact of AI. Yet doing so has been difficult. The field still lacks theories and scalable measurement frameworks for evaluating AI's generalization and using such measurements to systematically guide AI training. The current practice measuring generalization is to rely on a set of privately held-out "internal benchmarks," whose test data are explicitly excluded from the training corpus [22–24]. However, as frontier sytems are trained on increasingly diverse training corpora and environments, it has become much harder to define the boundaries between in-distribution and out-of-distribution since training data may contain highly similar or semantically equivalent instances of held-out evaluation tasks [25]; this limits scalability of these internal evaluations. Moreover, to the best of our knowledge, no established measurement tool exists for characterizing how an evaluation suite shapes human expectations about AI behavior nor the extent to which these expectations are broken or fulfilled in novel situations.

To address these issues, we introduce a formal framework for measuring *generalization surprise*, building on a *generalization model* using psychometric features that encodes human expectations of an AI's performance, given knowledge of that system's performance on a previous evaluation suite. This enables us to define (i) metrics for characterizing generalization of AI systems at scale; (ii) a fully-automated methodology for discovering surprising generalization failure modes in AI; and (iii) a diagnostic tool for assessing the ability of common evaluation suites to properly establish expectations for humans about AI's generalization. We analyze five families of frontier LLMs (GPT, Gemini, Qwen, Muse, Gemma) over a new evaluation suite that we introduce,

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termed 'GeneralBench,' covering 538 tasks from 85 benchmarks spanning a wide range of domains.

Our main findings are fourfold. First, we find that recent AI systems, though scoring similarly on standard benchmarks, can diverge notably in generalization surprise. Even closed frontier systems (e.g., GPT-6-Astra, Gemini-4-Argon) show substantial generalization surprise, especially for tasks outside the domain of quantitative reasoning, such as metacognitive assessment and written expression. Second, unlike standard measures that are commonly used on performance leaderboards, if we combine our notion of surprise with system error into a *generalization loss* metric, we can characterize an AI system's generalization while being resistant to myopic adversarial hill-climbing (e.g., training on test data) that inflate performance on individual benchmarks without improving generalization. Third, the trained generalization models can automatically surface many high-value and low-value items from existing benchmarks where system subjects surprisingly fail and succeed, respectively. Valuing benchmark items using our notion of surprise allows AI developers to systematically identify and patch generalization failure patterns in frontier AI systems, guide the creation of high-value future evaluations—we demonstrate this through introducing SurpriseBench, a simple benchmark where all evaluated frontier models score below 20% and 9% under pass^1 and pass^5 evaluations, respectively. Fourth, model-card evaluations of the latest frontier systems generally lead to substantially more surprises than our proposed approach when extrapolating model performance to unseen tasks, partly due to systematic coverage gaps. We encourage future AI developers to adopt our proposed metrics and methods for improving the training and evaluation suites of AI systems.

2 Modelling Generalization Surprise

Below we detail how we leverage a psychological latent space to capture the human-perceivable features of task items in a manner that is generalizable across arbitrary domains and items. Trained to approximate human generalization expectations given an AI system's past evaluation results, a generalization model takes each new item's latent vector as input to predict how well that AI system will perform on that item. This enables us to measure generalization at scale.

2.1 Properties of Human-centric Generalization

Ref. 26 proposed the universal law of generalization: a psychological space can be established for any set of stimuli (e.g., colors, shapes, sounds) by determining distances between the stimuli such that the probability that a response learned to any stimulus will generalize to any other is a monotonic function of the distance.

Inspired by Shepard's law, we assume the existence of a latent continuous metric space, $\mathcal{X} \subset \mathbb{R}^K$, with K dimensions that capture *human-perceived generalization patterns*. These K latent dimensions correspond to a set of *general* capabilities (e.g., quantitative reasoning, verbal comprehension, knowledge for formal sciences) required to solve arbitrary cognitive tasks. That is, given a latent space $\mathcal{X}$, any discrete task item i may be characterized by an *item-level demand vector* $\mathbf{x}_i \in \mathcal{X}$.

To define notions of human-centric generalizability within the latent space, we lean on two branches of literature related to Shepard's law of generalization. One branch comes from the machine learning community's notions of robustness [27–30] while another branch draws from item response theory studied in psychometrics [31, 32]. Unifying these two literatures under Shepard's law gives us two properties of human-centric generalization: (1) if two items are close in some latent space $\mathcal{X}$, an AI system's performance on those items should be similar; and (2) if an item is easier

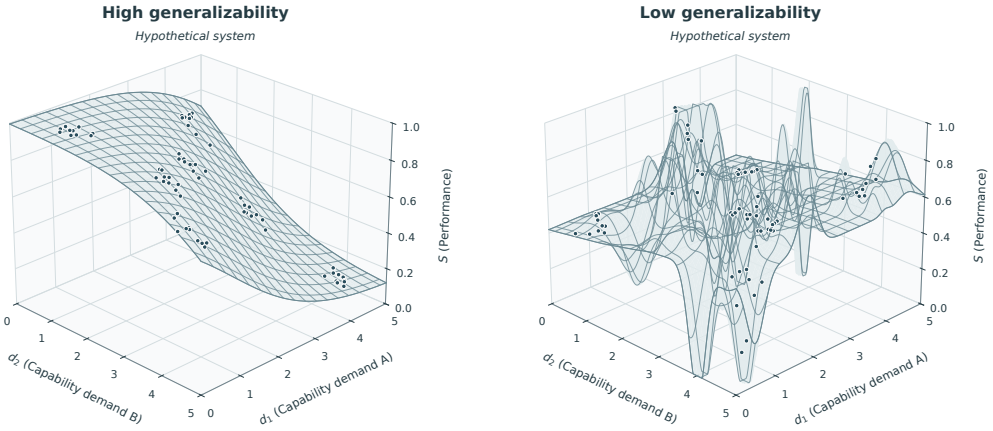


Fig. 1 Intuition of human-centric generalization. We show what high and low human-centric generalizability may look like in a hypothetical 2-D latent space $\mathcal{X} \in [0,5]^2$, where dimensions reflect demand for specific capabilities and dots correspond to task items. (Left) A hypothetical system with high generalizability that exhibits a smooth performance surface. (Right) A system with low generalizability, characterized by a spiky performance surface where negligible shifts in demand result in unpredictable performance swings.

than another item, a subject should score higher on the easier item. For intuition, Figure 1 visualizes the two generalization properties for an AI system with high human-centric generalization and one without. For a more formal and extensive treatment of human-centric generalization, see Appendix A.1.

2.2 Model of Human-centric Generalization

For simplicity, we implement our generalization model as a binary classifier, such that $S(M,i) \in \{0,1\}$ where 1 indicates model success, but our framework generalizes to multi-class scoring functions. To satisfy both properties of human-centric generalization, the generalization model p_{θ_M} is restricted to a logistic regression classifier with non-positive coefficients. The logistic classifier approximates property (1) while the non-positive coefficients satisfy property (2).

Formally, our generalization model p_{θ_M}, fed with item-level demand vectors $\mathbf{x}$, is as follows:

$$p_{\theta_M}(y = 1 \mid \mathbf{x}) \;=\; \sigma(\theta_0 + \boldsymbol{\theta}^\top \mathbf{x}), \qquad \boldsymbol{\theta} \in \mathbb{R}^K_{\leq 0}. \tag{1}$$

This sigmoid function follows the standard parametric form used in psychometrics for evaluating human and animal intelligence [33–35], and also recently transferred to AI capability evaluation [4, 32, 36].

2.3 Defining Generalization Surprise

Let $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{ref}}$ denote sets of training and held-out reference items, respectively. Below, j and i correspond to item objects (i.e., prompts in natural language) within $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{ref}}$, respectively, while $\mathbf{x}_j$ and $\mathbf{x}_i$ denote the item-level demand vectors of j and i. And, $S(M,i)$ corresponds to the AI system M's performance score on item i. With this, we define the history of item scores for system M be

$$\mathcal{H}_M = \{(\mathbf{x}_j, S(M,j)) \mid j \in \mathcal{D}_{\text{train}}\} \tag{2}$$

We train the generalization model p_{θ_M} to predict how AI system M performs on each item from $\mathcal{D}_{\text{train}}$ by giving the history $\mathcal{H}_M$ as input, resulting in a $p_{\widehat{\theta}_M}$ with fitted parameters. Then, each held-out item i has a *generalization surprise* score $\ell_{M,i}$ that reflects the extent to which M's actual performance deviates from the expectation of $p_{\widehat{\theta}_M}$:[1]

$$\ell_{M,i} = S(M,i) - p_{\widehat{\theta}_M}(y = 1 \mid \mathbf{x}_i), \qquad i \in \mathcal{D}_{\text{ref}} \tag{3}$$

This can be useful to support sample-efficient AI training by allowing us to scalably locate high-value and low-value items, where M surprisingly fails and succeeds, respectively. Items with negative $\ell_{M,i}$ (surprising failures) reveal critical generalization gaps that should be prioritized during training, while items with positive $\ell_{M,i}$ (surprising successes) disclose areas where the AI scores surprisingly higher than the expectation formed by previous evaluations.

Intuitively, if system M generalized perfectly to match human expectations, ℓ_{M_i} would always be zero, but non-zero otherwise. If we aggregate $\ell_{M,i}$ over all test items $i \in \mathcal{D}_{\text{ref}}$, we get a global measure of generalization surprise of M[2] :

$$\text{SURPRISE}(M; \mathcal{D}_{\text{ref}}) = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{i \in \mathcal{D}_{\text{ref}}} \ell_{M,i}^2 \tag{4}$$

Note that SURPRISE is distinct from standard performance scores like error rate because reduced (average) error does not necessarily lead to reduced generalization surprise (see Figure 2).

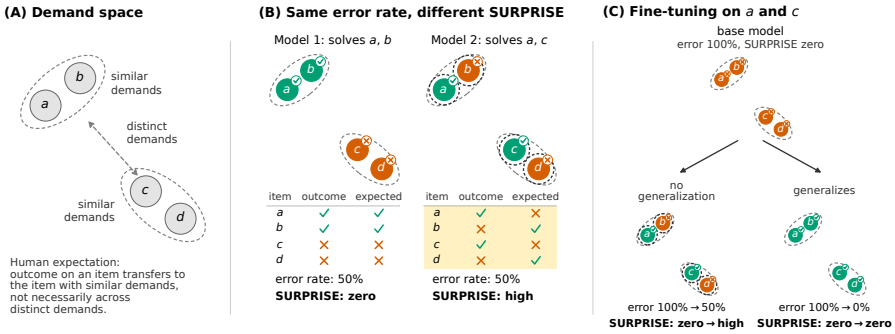


Fig. 2 The difference between generalization surprise and error rate, illustrated in a hypothetical 2D demand space. (A) Four items whose demands form two clusters; a human expects performance to transfer within a cluster ($a \leftrightarrow b$, $c \leftrightarrow d$) but not across clusters. (B) Two AI systems with same error rate but divergent SURPRISE. (C) Fine-tuning a base AI system on a and c reduces the error, but only generalizable fine-tuning keeps SURPRISE at zero.

2.4 Human-centric Generalization Loss

SURPRISE is useful for quantifying overall surprise over $\mathcal{D}_{\text{ref}}$, but it is not equivalent to measuring generalization. For example, when an AI system always fails, its behavior is perfectly predictable and therefore the system has zero SURPRISE (See Figure 2c). To characterize desirable generalization, we instead combine both the AI error and generalization surprise as follows:

[1]This can be thought of as similar to the notion of surprisal by Ref. 37.
[2]We square $\ell_{M,i}$ so that SURPRISE equals the Brier score [38], a proper scoring rule quantifying the mean squared gap between the generalization model's predictions and M's actual performance.

$$\text{GENLOSS}(M; \mathcal{D}_{\text{ref}}) = \underbrace{1 - \bar{S}_M}_{\text{error penalty}} + \alpha \underbrace{\left|\bar{S}_M - \bar{P}_M\right|}_{\text{surprise penalty}}, \tag{5}$$

where $\bar{S}_M$ denotes the observed average performance of M over $\mathcal{D}_{\text{ref}}$, and $\bar{P}_M$ denotes its predicted average performance over $\mathcal{D}_{\text{ref}}$. As shown in Figure 3, GENLOSS nullifies gains from above-expected performance (i.e., surprising successes) and penalizes below-expected performance (i.e., generalization failures). Throughout this work, we set $\alpha = 1$. Both penalty terms lie in $[0, 1]$ when $S(\cdot, \cdot)$ is accuracy, so GENLOSS $\in [0, 2]$, with larger values indicating higher error or a larger generalization gap.

GENLOSS represents a *generalizable optimization target*, which can only be minimized if the system improves its performance while transferring such gains in other untrained tasks used to construct the generalization model. To evaluate overall system intelligence, $\mathcal{D}_{\text{ref}}$ should contain diverse cognitive tasks. However, if one wishes to characterize an AI system's generalization for a specific set of items (e.g., in a particular domain like math), they may modify $\mathcal{D}_{\text{ref}}$ to contain a set of items reflective of the set in question that matches their downstream interests.

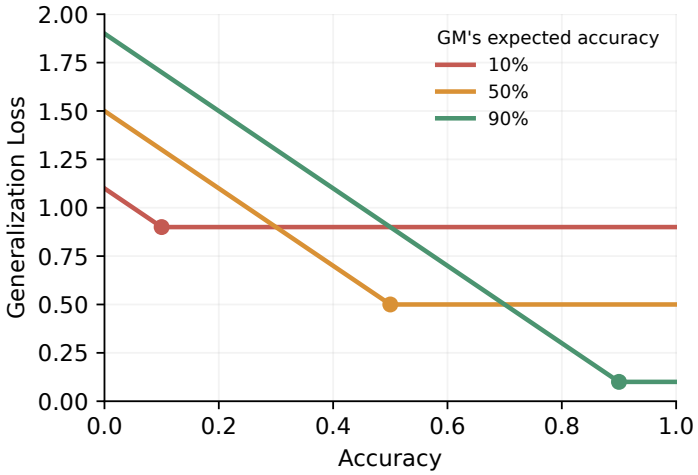


Fig. 3 The connection between generalization loss and accuracy. We plot GENLOSS for three fixed generalization model expectations and show the effects of increasing accuracy under fixed generalization model (GM) expectation. No reward is given for inflated performance and the AI system is penalized when it underperforms the generalization model's expectation.

3 Experimental Design

3.1 Instantiating the Latent Demand Space

Our framework requires instantiating the latent space $\mathcal{X}$. To achieve this, we explore three taxonomies of general cognitive capabilities and skills in the literature: ADeLe v.1.0 [4], O*Net [39] and OECD capability indicators [40]. Appendix A.3 elaborates on these three taxonomies and a preliminary large-scale analysis with across 110 latent dimensions. There, ADeLe achieved the best predictive power among the three and thus will be the main taxonomy used in our subsequent analyses. Still, O*Net also performed comparably compared to ADeLe and thus may be considered as an alternative (or complemantary) taxonomy for future work using our framework.

In summary, ADeLe v.1.0 contains 19 dimensions of cognitive capabilities, knowledge and extraneous factors inspired by the literature in human psychology, comparative

cognition and artificial intelligence. These include verbal comprehension and expression, logical and quantitative reasoning, mind modeling and social recognition, attention and scan, distinct branches of knowledge, among others. We adopt the subset of 16 dimensions from ADeLe that are supported by literature documenting their presence in humans, while excluding extraneous factors. For simplicity, we refer to this 16-dimensional subset as ADeLe throughout this work. We recommend checking [4] for further details. Each capability dimension is defined by an expert-validated rubric, ranging from 0 to 5—zero represents the absence of capability demand while values 1-5 represent increasing demand levels for that respective capability, with $\mathcal{X} \in [0, 5]$[19].

3.2 Data Curation and Model Subjects

We collected 85 LLM benchmarks with 538 task categories from major AI conference proceedings, leading scientific journals, and model cards of every studied LLM subject[3]. We exclude agentic tasks that cannot be evaluated on non-agentic LLMs, which leave for future work. We describe the curation details and these benchmarks in Appendix A.4. For improving evaluation efficiency while keeping diversity, we apply medoid-based sampling [42, 43] over item embeddings when curating benchmarks. For each task, we sample up to 50 items by embedding each item with bge-large-en-v1.5 [44], running k-means with $k = \min(50, \text{task size})$, and selecting medoids [42, 43]. This led to a total of 21k items. Collectively, we term this new collection of diverse items "GeneralBench."

The main pool of subjects comprises 12 LLMs from five families (i.e. GPT, Gemini, Qwen, Gemma, Muse). Six are proprietary systems from OpenAI and Google: GPT-6-Astra, GPT-6.1-Sol, GPT-5.6-Terra, and GPT-6-Luna [6, 45]; and Gemini-4-Argon, Gemini-3.1-Pro and Gemini-3.8-Flash [8, 46, 47]. Other five are open-weight systems from Alibaba, Google, and Meta: Qwen3.6-27B, Qwen3.8-27B, and Qwen3.8-Max [48, 49]; Gemma-4-31B [50]; and Muse-Glimmer-30B [51]. Unless otherwise noticed, our main analyses will focus on these main 12 subjects. Appendix A.5 details the inference and grading setups.

3.3 Demand Annotations and Generalization Models

We use GPT-5.6-Terra [45] at low reasoning effort as the demand annotator, using the same prompt and rubrics as Ref. 4 to annotate the demands of items in GeneralBench. This led to a total of 340K annotations. These annotated demands are fed into the generalization model. Unless otherwise specified, we use leave-one-benchmark-out cross-validation (LOBOCV) across 85 benchmarks for each AI system M. In each fold, we fit a generalization model on items from 84 benchmarks ($\mathcal{D}_{\text{train}}$) and evaluate it on items from the held-out benchmark ($\mathcal{D}_{\text{ref}}$). We then macro-average the resulting held-out results across benchmarks to compute metrics like SURPRISE and GENLOSS. We obtain $R = 1000$ bootstrapping replicates for quantifying epistemic uncertainty. Importantly, through a human study in Appendix A.6, we show that the trained generalization models indeed capture well human-like generalization patterns.

4 Experimental Analyses

4.1 Generalization Surprise

Figure 4 demonstrates that generally the largest close-weight frontier systems not only improve in accuracy but also moderately reduce overall surprise in comparison with earlier counterparts or smaller systems from the same family. In contrast, for less

[3] All LLM benchmarks of the LLM subjets' model cards are included, with the only exception of CritPt [41] because its private evaluation only returns aggregate metrics and not per-item scores.

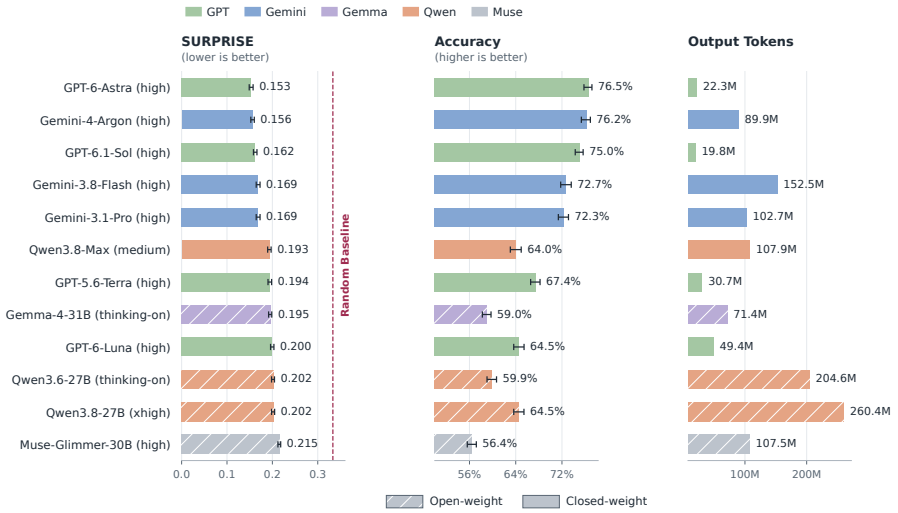


Fig. 4 Generalization surprise and accuracy of studied LLM subjects on our (non-agentic) GeneralBench. We macro-average SURPRISE under LOBOCV, macro-average accuracy, and total output tokens. Bars are medians and 95% percentile intervals over 1000 bootstrap replicas. Token usage varies across systems, so the rankings should be interpreted with caution.

capable AI systems—mostly open-weight systems except for GPT-5.6-Luna—higher accuracy scores do not often lead to lower generalization surprise. We conjecture two complementary explanations: (1) small-sized systems generalize worse and (2) open-weight AI developers are less-resourced and thus more prone to data leakage issues or pressured to train systems on headline benchmarks (i.e., "benchmaxxing") reported by their own model cards [25, 52–54].

Despite these improvements, the SURPRISE scores are still large even for the best AI systems relative to a uniform random baseline, indicating substantial room for improvement toward building AI systems that are not only capable but also free from surprising generalization failures.

4.2 Generalization Loss

We characterize model generalization in detail using our generalization loss metric (Methods 2.4) in Figure 5. We observe two key takeaways. First, since GENLOSS is robust against adversarial hill-climbing that inflate performance on individual benchmarks without generalizing to other unseen benchmarks (see Figure 6 and Appendix A.2), the generalization improvements of latest frontier systems seem genuine. Nonetheless, a huge variance between domains still exists. For example, the systems generalize much better tasks that demand high quantitative reasoning, spatio-physical reasoning, and knowledge in formal sciences (e.g. math, computer science)[4]. In contrast, domains like verbal expression, calibrating knowns and unknowns, and mind modeling manifest much more limited progress, revealing areas that may warrant prioritization in AI training. This divergence may be largely attributed to the latter domains being difficult to verify, corresponding to underlying dimensions that materially shape intelligent behavior yet are hard to reliably infer from behavior alone. Because of this, such dimensions have been relatively under-represented in AI training data, and thus are poorly instilled during AI training [58].

[4]This seems aligned with the recent focus of frontier labs on math and coding [55–57] since they are more verifiable

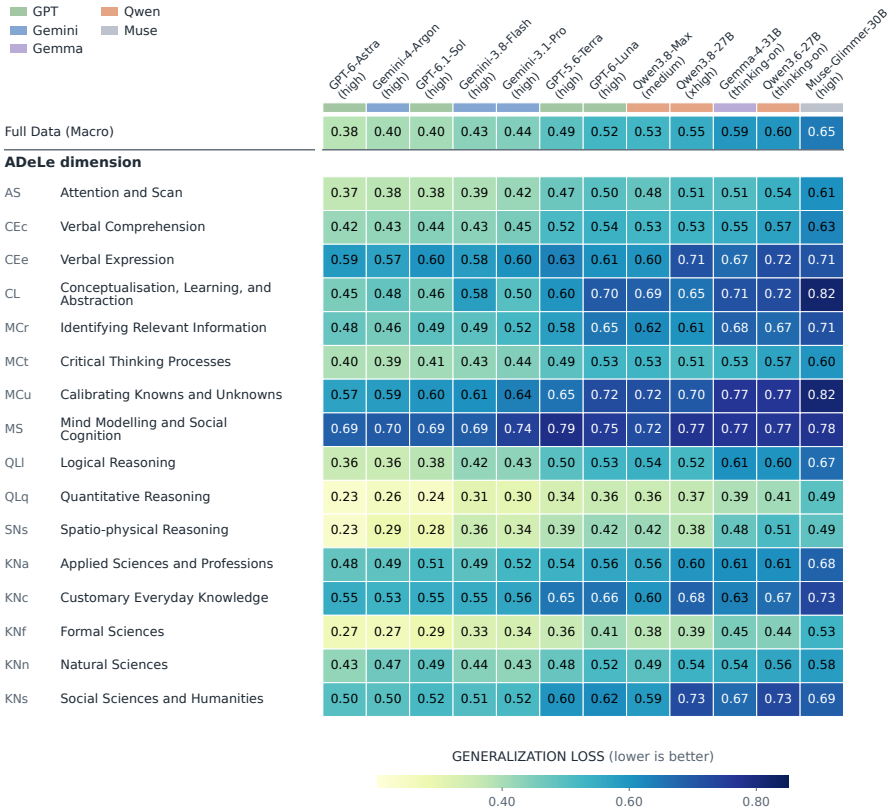


Fig. 5 Generalization loss disaggregated by ADeLe dimensions. The top row is the (total) macro-averaged GENLOSS under LOBOCV. Other domain-specific rows use dominant slicing following [4]: When evaluating a given domain or dimension, keep the items whose demand on that dimension is the highest among other dimensions. The cell numbers are still macro-average across benchmarks, but the calculation seeks robustness by excluding noisy benchmarks with less than ten items.

These findings can be supported by an further analysis in Appendix A.7, where we show that AI systems tend to reduce their errors without notable reductions in SURPRISE in those (aforementioned) domains with limited progress, unlike other more verifiable domains. Furthermore, Appendix A.8 provides a scaling analysis by scaling up system sizes and test time compute of multiple LLM families (i.e. Qwen, GPT or Gemini). In summary, larger system size clearly improves generalization (i.e. reduced GENLOSS), while more test time compute also leads to moderately improved generalization but much more limited in non-reasoning-heavy tasks.

4.3 Surprising Generalization Failures

To address the generalization failures discovered in our previous analyses, it is essential to understand and patch cases where frontier AI systems surprisingly fail. In a preliminary analysis, we found three broad clusters of failures: Low-quality data (e.g., ambiguity or errors in the prompts or gold answers), imperfect graders, and genuine system failures. This guided us to both significantly reduce the number of low-quality items and improve the grading quality across nearly all benchmarks that used programmatic scoring (see details in Appendix A.9), before conducting our main analysis. This shows another usefulness our framework: Flagging issues of task validity and outcome validity [59] to enhance benchmark rigor. In Appendix A.9, we additionally provide quantitative and qualitative analyses of genuine surprising failures (e.g. GPT-6-Astra

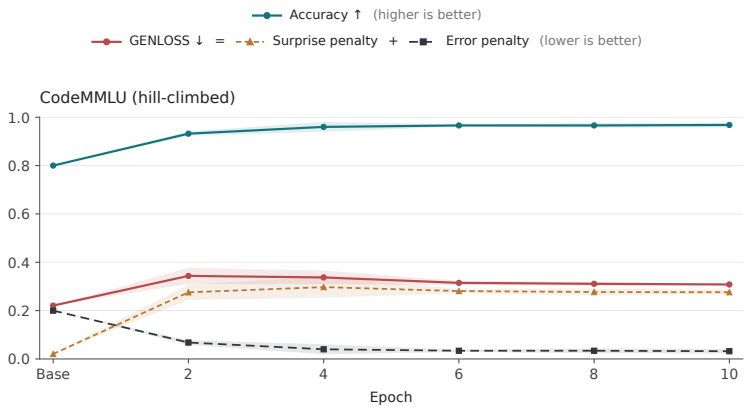


Fig. 6 Despite improved accuracy, generalization loss resists to go down when Gemma-4-26B-A4B is trained on CodeMMLU's gold answers over 10 epochs without generalizing to held-out benchmarks, thanks to increasing 'surprise penalty'. The generalization model is trained on the results of held-out benchmarks to predict the hill-climbed benchmark (CodeMMLU). The results are averaged across five random seeds, with 95% Wald CI. Appendix A.2 and Figure 10 detail this further alongside various examples of "benchmaxxing" for domains beyond coding.

fails in a trivial short-sequence calculation, GPT-5.6-Sol gets wrong on a basic logical reasoning problem, Qwen3.8-Max producing garbled text instead of an actual answer) found in our suite of 85 benchmarks.

Beyond surfacing failures in the collected benchmarks, our framework also allows for creating new, surprising failures. We use GPT-6-Astra with agentic scaffolding and minimal human steering to efficiently generate a synthetic dataset of 200 problems across diverse failure modes, which we call 'SurpriseBench,' where most of our studied LLM subjects fail (i.e. these failure modes are transferable across systems[5]). Here, all studied LLM subjects score below 20% and 9% under pass^1 and pass^5 evaluations, respectively (Figure 7). Appendix A.10 further details the specific failure modes alongside qualitative examples such as fabricating a transcript for a video that was never attached, silently overlooking contradictory user statements when drafting factual reports, reporting results contradicted by their own verification code, altering data explicitly requested to be copied verbatim, and producing strongly biased outputs when explicitly asked for uniform random sampling.

To inform AI progress in terms of reducing surprising failures, we open source 100 problems of our SurpriseBench while holding the other 100 problems on private. These initial results, based on our notion of generalization surprise, shed light on the promise of automatically surfacing generalization failure patterns for guiding the improvement of AI generalization at scale. We leave further exploration of these exciting directions to future work.

4.4 Inadequacy of Model Cards

Evaluation suites in model cards are key reference points for researchers, users, and policymakers to understand AI's capabilities and generalization [5]. However, to our knowledge, no prior work has evaluated whether model cards actually set adequate expectations to minimize human surprise. To evaluate the quality of the model cards, we conduct two experiments: (1) we compare the SURPRISE scores between the generalization model trained on GeneralBench and the generalization model trained

[5]Transferrable failure modes were found to be much harder to find than system-specific failure modes, likely because of the differences in training recipes, which may induce distinct unintended behavior for individual LLMs to learn.

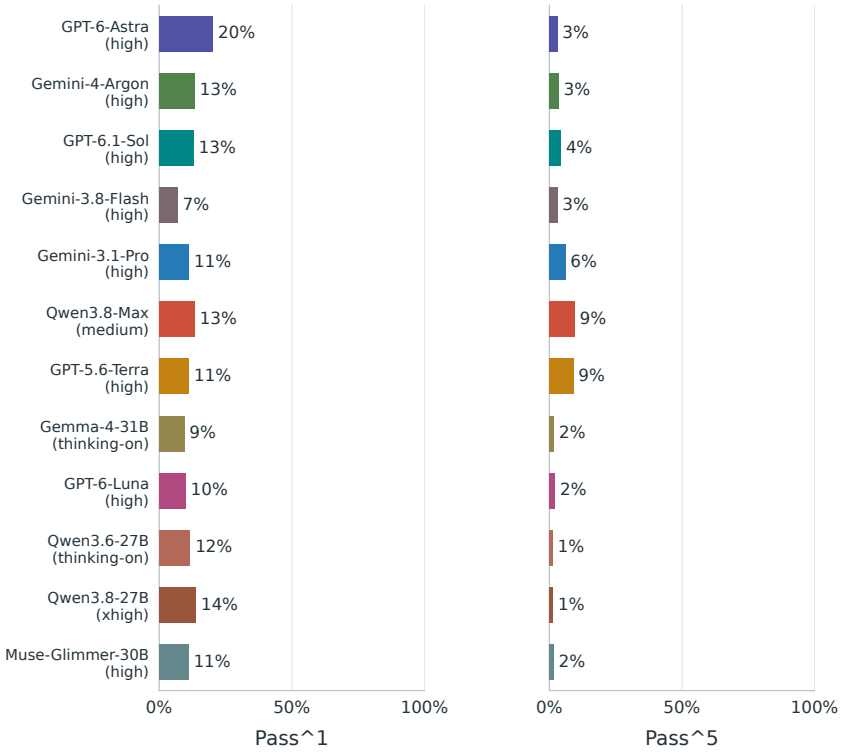


Fig. 7 All studied LLM subjects score poorly under pass^1 and pass^5 evaluations on our SurpriseBench. All 200 new problems from SurpriseBench have $\ell_{M,i} < -0.6$ for GPT-6-Astra and Gemini-4-Argon (i.e., conditioned on their previous performance on the 85 collected benchmarks of our main analysis, the generalization model would predict that these LLMs are likely to succeed, but they actually failed in at least one out of five trials during the data generation phase.).

on model card benchmarks and (2) we compare the coverage and diversity of demands between our GeneralBench and model-card evaluation suites. We collected all LLM benchmarks from the model cards used by each tested LLM subject (described in Methods 3.2).

Figure 8 reveals that SURPRISE is much higher conditioned on model-card evaluations in comparison with (an equal-sized subset of) GeneralBench, under a fair comparison, only except for Qwen3.6-27B where SURPRISE is similar. We believe these high SURPRISE scores of model cards can be largely explained by their lack of coverage in the latent demand space: Figure 9 shows that, compared with GeneralBench, model card items taken cumulatively cover generally much less of the latent space—this makes it harder to extrapolate AI performance to diverse unseen scenarios.

Taken as a whole, our findings illustrate that current model cards fail to set proper expectations for AI system performance. To address this, future model cards should adopt our framework of generalization surprise to guide evaluation construction, such as improving the evaluation suite's diversity, or any other resorts that help humans to set reliable expectations.

5 Related Work

A large body of work has examined ways to evaluate model generalization or non-generalizable behavior [7, 25, 30, 45, 52, 60–65] and data valuation [66–73]. Going beyond these prior related work, our work introduces a formal notion of human-centric generalization surprise, allowing us to scalably evaluate generalization over a latent

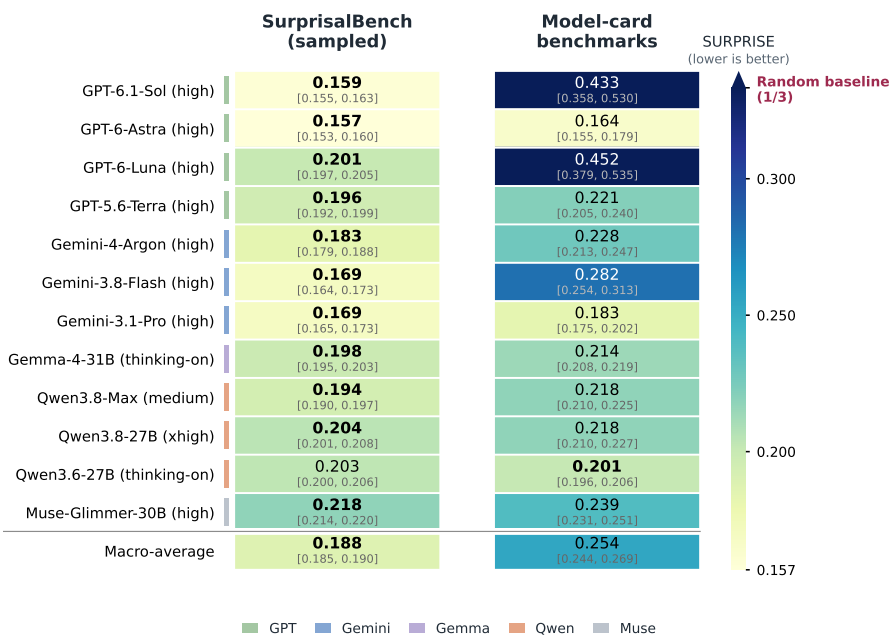


Fig. 8 The generalization model trained on GeneralBench (under LOBOCV) yields lower SURPRISE **than the generalization model trained only on model-card benchmarks**, when predicting (held-out) non-model-card benchmarks in GeneralBench. For a fair comparison, all cognitive models are trained with same number of samples. Cell values are medians accompanied by 95% percentile intervals over 1000 bootstrap replicas.



Fig. 9 LLM benchmarks of recent model cards cover only limited regions of the latent demand space, based on a 2-D t-SNE projection of GeneralBench data, fed with 19 ADeLe demand features using `scikit-learn`. The red dots represents items in our GeneralBench but not in the subject's model card, while the green dots correspond to items that appear in both.

demand space covering diverse cognitive domains, surface high-surprise data that are informative for improving evaluation quality and AI training, and monitor the quality of model card evaluations. Building on generalization surprise and error metrics, we additionally introduce another metric (generalization loss) for characterizing AI's human-centric generalization at scale, fighting against score-hacking. Beyond technical and empirical contributions, we argue that future model card evaluation suites should establish proper generalization expectations without surprises, and training recipes of frontier AI systems should optimize for generalizable optimization targets like GENLOSS to assess performance transfers from internal benchmarks to real-world deployments. For a prolonged discussion of all relevant and peripherally related work, see Appendix A.11.

6 Discussion

AI systems are now matching or outperforming the leading human experts on specific subtasks across many domains [17, 74–78], but still manifest myriad surprising failures—often with severe consequences, especially in high-stakes applications [10–14, 16–18, 60, 79, 80]. This lack of human-centric generalization places a huge burden on users and policymakers to determine when and how much these systems should be trusted, undermining the positive societal impact that AI could otherwise deliver. Identifying and patching such surprising failures to better align machine behavior with human expectations, and reporting evaluations that set proper expectations for humans, are hence of great importance. Nonetheless, these two objectives have been challenging to achieve.

We address these by proposing a generalization model that scalably encodes human expectations about how performance observed on previous evaluations should transfer to new tasks, spanning 85 benchmarks across a wide range of cognitive tasks and mainstream AI families. This allows us to design novel metrics for tracking genuine generalization improvements as well as systematically characterizing where AI systems generalize as expected and where they do not. These can, in turn, support the automatic identification and patching of generalization failures in future AI systems, thereby improving their generalizability and reliability. Beyond the system side, our framework is the first of its kind for quantifying how adequately an evaluation suite shapes human expectations about an AI system's broader behavior. This has enabled us to reveal systematic coverage issues in current evaluation suites accompanying AI releases for establishing reliable human expectations about system behavior. This, too, needs to be overhauled, so AI evaluations can adequately inform users and avoid undesirable surprises, especially in high-stake scenarios. Taken together, we encourage AI developers to adopt our methodology for systematically understanding and fixing generalization failures, while designing compact evaluation suites that efficiently communicate what new AI systems can—and cannot—be expected to do.

Our work is not without limitations. First, due to prohibitive cost of collecting several hundreds of diverse tasks, we start our framework analyzing LLMs on non-agentic settings. However, this can be straightforwardly adapted to AI agents and embodied systems in future work, provided the presence of a generalization model that captures human generalization expectation. Future efforts building on our framework are needed to create a holistic benchmark suite for AI agents, possibly hidden and dynamic, to build a hard-to-game third-party leaderboard that can reliably monitor both the generalization of new AI systems and the quality of model cards, to better inform broader public. Future work should also leverage our human-aligned notion of failure surprise to improve AI systems, because failure surprise may serve as a valuable reward signal for discovering valuable new data: the more a failure violates reasonable

human expectations with respect to past evaluations, the more informative the example is to improve the system.

The concepts and methodology we have introduced here showcase the successful adoption of a human-centric approach on scalably characterizing generalization surprises, integrating perspectives from multiple disciplines. As AI systems continue to match or outperform the best domain experts on arbitrary problems, characterizing human-centric generalization and reducing surprising failures will be central to guiding research and regulations toward building reliable and safe general intelligence. This work represents a cornerstone for developing measurement standards and AI systems suited to that future.

Data and Code Availability. To foster research and policymaking, we open-source our data, code, results, and an interactive experiment viewer: https://generalization-surprise.pages.dev.

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A Supplementary Information

A.1 Formalizing Human-centric Generalization Definitions

To define notions of human-centric generalizability, we lean on two branches of literature related to Shepard's law of generalization. One branch comes from the machine learning community on neighborhood invariance [28, 30], while another branch corresponds to item response theory [31, 32]. We bring these two schools of thoughts together under an unified view of Shepard's law, constructing a *generalization model* to capture the two aforementioned human-centric generalization properties (Definition 1 and Definition 2).

Given a latent space $\mathcal{X}$ and a reference distribution $\mathcal{D}_{\text{ref}}$ over $\mathcal{X}$, we can define two human-centric generalization properties (Definition 1 and Definition 2) expected from AI systems with perfect generalization. The first property is inspired by prior ML research on robustness, specifically Theorem 6 of the work by Ref. 28 and neighborhood invariance for ML classifiers [30]. Therefore, the first property advocates that general systems should yield similar scores when transferring to similar, unseen tasks:

Definition 1. Let $(\mathcal{X}, d)$ denote a latent demand space with distance d, and let $S(M, i)$ be the system M's performance score on item i. A system M is said to exhibit perfect *performance neighborhood invariance* if, for all $\epsilon > 0$, there exists a $\delta > 0$ independent of $x \in \mathcal{D}_{\text{ref}}$ such that if:

$$d(\mathbf{x}_i, \mathbf{x}_j) < \delta \implies |S(M, i) - S(M, j)| < \epsilon, \quad \forall \mathbf{x}_i, \mathbf{x}_j \in \mathcal{D}_{\text{ref}} \tag{6}$$

For example, because humans perceive the distance d between two high-school algebra problems as small, a general system is naturally expected to yield similar performance on both tasks. Similarly, two versions of the same high-school algebra problem that differ only slightly in their natural language formulation can be viewed as variations with zero distance in $\mathcal{X}$, and hence a system with perfect generalization is expected to perform equally.

The second generalization property is inspired by the literature of item response theory, evidencing that human performance decreases monotonically as task difficulty increases [31, 32]. Building on this, we assert that an AI system with perfect human-centric generalization should be capable of solving not only challenging (high demand levels) problems for a given domain, but also excelling at simpler (lower demand levels) problems within the same task domain. We formalize this idea as performance transition monotonicity in Definition 2.

Definition 2. Let $(\mathcal{X}, \preceq)$ be a partially ordered latent demand space, where the partial order $\preceq$ is defined component-wise: $\mathbf{x}_i \preceq \mathbf{x}_j$ if and only if $x_{i,k} \preceq x_{j,k}$ for all latent dimensions $k \in \{1, \ldots, K\}$. A system M is said to have perfect *performance transition monotonicity* if:

$$\mathbf{x}_i \preceq \mathbf{x}_j \implies S(M,i) \geq S(M,j), \quad \forall \mathbf{x}_i, \mathbf{x}_j \in \mathcal{D}_{\text{ref}} \tag{7}$$

For example, a system that achieves a gold-medal score in the IMO but fails at elementary math constitutes a non-monotonic transition since the latter is usually a pre-requisite to solve IMO problems.

For intuition, Figure 1 visualizes the two generalization properties for a system with perfect generalization and one without.

A.2 Differentiating Generalization Loss from Error Scores

The key distinction between our GENLOSS measure (Eq. 5) and standard error scores is that error scores can be reduced without necessarily reducing generalization loss. To further illustrate this distinction, we conduct a detailed ablation study (Figure 10) over four domains[6], showing how AI's performance scores can increase overall without reduction in generalization loss (and sometimes even increasing generalization loss). More concretely, we adversarial hill-climb on four benchmarks from distinct domains via parameter-efficient supervised fine-tuning (SFT) on Gemma-4-26B-A4B with LoRA over 10 epochs. We then measured the system's accuracy and GENLOSS on the hill-climbed benchmark and 19 held-out benchmarks[7] (Figure 10), obtaining supporting findings. For the contaminated benchmark, the LLM's accuracy increased for the hill-climbed benchmark (i.e., reduced error penalty), but the generalization loss also increases because the surprise penalty—one of the two components forming the generalization loss—substantially increases too. This is expected, as the system's inflated performance failed to generalize to new tasks. This can be exemplified by the right hand side of Figure 10, where we see that system accuracy on held-out benchmarks decreases, and generalization loss generally either stays flat or increases, indicating a worse overall generalization gap between the 20 benchmarks.

[6]This includes coding (CodeMMLU), scientific question answering (GPQA-Diamond), long-context understanding (LongBench-v2), and logical reasoning (BBEH).

[7]We use 19 held-out benchmarks to be frugal with compute.

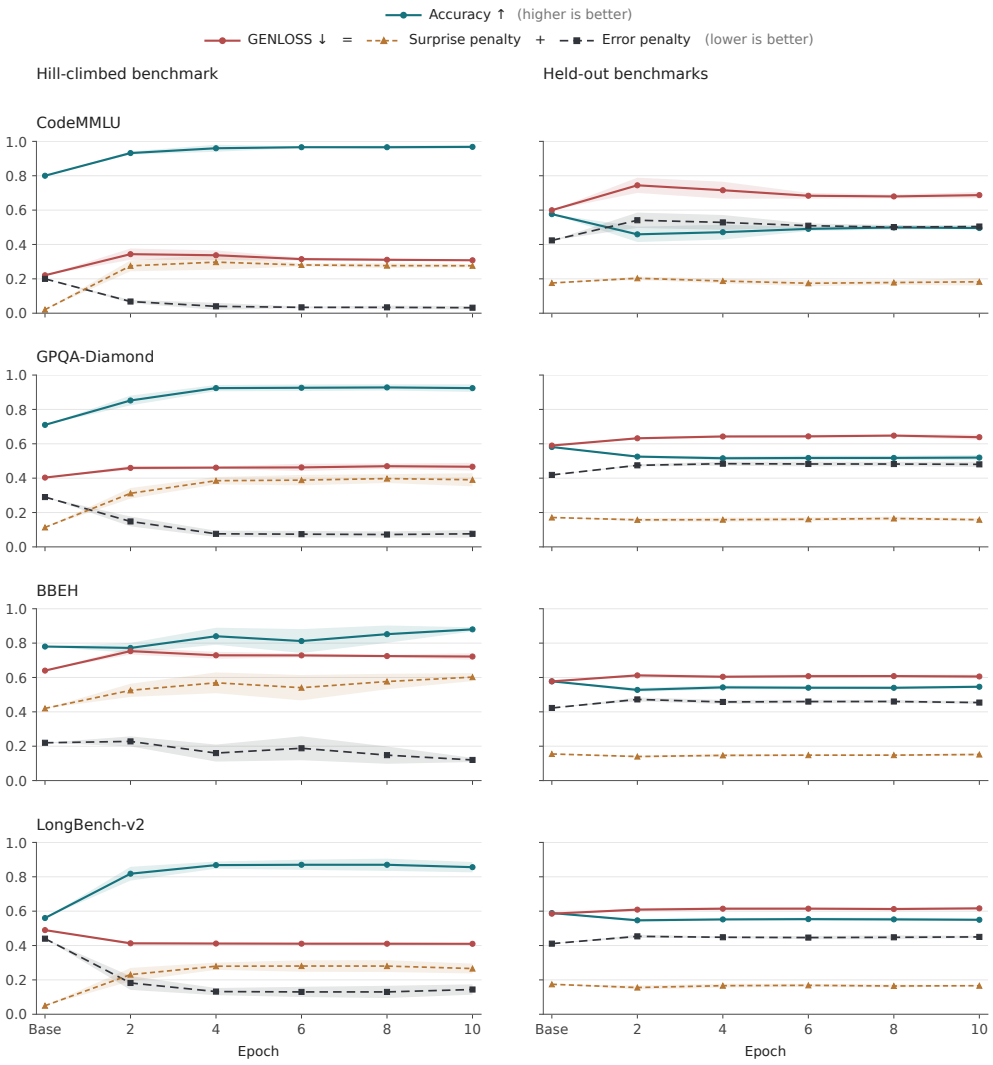


Fig. 10 Despite improved accuracy, generalization loss resists to go down when Gemma-4-26B-A4B is trained on various benchmarks' gold answers over 10 epochs without generalizing to held-out benchmarks, thanks to increasing 'surprise penalty'. The generalization model is trained on the results from 19 held-out benchmarks to predict the given hill-climbed benchmark. The results are averaged across five random seeds, with 95% Wald CI.

A.3 Detailed Description of the Three Cognitive Taxonomies

This appendix further details the three taxonomies of cognitive capabilities and skills, and how we adopted them to instantiate the latent space $\mathcal{X}$ for our analyses.

ADeLe v1.0 [4] carves the space of LLM capabilities into a hierarchical set of 19 general dimensions, inspired by the literature in human psychology, comparative cognition and artificial intelligence. Each dimension is defined with a expert-derived rubric, readily allowing for item-level annotations of demand levels from 0 to 5 using LLM judges. These 19 dimensions are organized into three groups: 11 cognitive abilities (e.g. Logical Reasoning, Metacognition, Comprehension, Mind Modelling), 5 knowledge dimensions (e.g. Formal Sciences, Social Sciences and Humanities), and 3 control factors that affect task difficulty independent of cognitive demands (i.e. Volume, Atypicality

and Unguessability). Every dimension is defined via an expert-derived rubric, with a quantitative range between level 0 and level 5, where 0 represents absence of demand of the respective capability, and values 1-5 represent increasing demand levels. We make use of a subset of 16 general dimensions (excluding extraneous factors) for instantiating the latent space $\mathcal{X}$, which are supported by literature documenting their presence in humans.

O*Net [39], which stands for Occupational Information Network (O*NET) 30.1 database , containing expert-derived dimensions and levels, where the K dimensions and their respective magnitudes are directly annotated by human job analysts and subject matter experts. Concretely, we instantiate $\mathcal{X}$ at the Generalized Work Activity (GWA) level, which defines 41 general activity categories—such as *Analyzing Data or Information* or *Interacting With Computers*—shared across 923 recorded occupations. Under this framework, each occupation is characterized by the specific subset of GWAs required for the role. This occupation-activity mapping allows the latent demand space to be populated by capability dimensions defined within the O*NET Content Model. More specifically, we incorporate 21 *Abilities* (e.g., Deductive Reasoning, Mathematical Reasoning) and 31 *Skills* (e.g., Programming, Critical Thinking), which are annotated at the activity level, applicable to LLMs. Additionally, we include 33 *Knowledge* domains (e.g., Mathematics, Engineering and Technology, Law and Government), which are annotated at the occupation level due to the lack of GWA-level data on these knowledge dimensions in O*Net. Together, these constitute $K = 85$ capability dimensions. The demand values x_{ik} are derived from O*NET's *Level* scores, an interval scale ranging from 0.0 to 7.0. These values are human-annotated based on behavioral anchors; for example, a level of 2.2 in *Written Expression* might represent writing a simple grocery list, whereas a 6.1 represents drafting a complex technical manual. Since O*NET annotations are provided at the activity or occupation level rather than the individual task-item level, we randomly sample 1000 occupation-activity pairs from the curated O*NET-GWA matrix to serve as in-context demonstrations for an LLM judge. Each demonstration consists of a natural language description of the work activity within its occupational context, paired with its associated $K = 85$ continuous-valued demand vector. By providing these few-shot examples, the LLM judge may be calibrated to the O*NET scoring logic, allowing it to project arbitrary benchmark items into $\mathcal{X}$ by estimating their latent demands relative to the human-anchored proficiency levels established by the O*NET subject matter experts.

OECD Capability Indicators [40] is a framework developed to systematically measure AI capabilities against human abilities. The OECD AI Capability Indicators define $K = 9$ broad capability dimensions (e.g. Language, Social Interaction, Problem Solving, Knowledge, Learning and Memory) that are applicable to LLMs, which we use to instantiate $\mathcal{X}$. Each dimension is evaluated on a five-level scale, where level 1 represents rudimentary capabilities and level 5 indicates the full range of human abilities. These indicators were developed with 50 AI researchers and psychologists. While the OECD originally designed these indicators to assess the capability levels of AI systems, we adapt them for item-level task demand annotation: Given a task item and an OECD dimension k with its associated rubric, we prompt an LLM judge to estimate the minimum capability level required along dimension k to successfully complete the task, assuming all other dimensions are not the bottleneck.

Table 1 shows that using the ADeLe taxonomy to train the cognitive model

A.4 Benchmark Details

Here we presents further details of the selected benchmarks. The 85 benchmarks are detailed in Table 2. These are collected from major AI conference proceedings,

Table 1 SURPRISE (Macro-average) across taxonomies. Values are reported with 95% bootstrap confidence intervals. The best-performing taxonomy for each system is bolded.

Model	ADELE	OECD	O*NET
gpt-5.6-sol-high	**0.166 [0.151, 0.181]**	0.184 [0.170, 0.198]	0.173 [0.159, 0.189]
gpt-5.6-terra-high	**0.183 [0.171, 0.196]**	0.194 [0.183, 0.206]	0.187 [0.175, 0.200]
gpt-5.6-luna-high	**0.185 [0.173, 0.199]**	0.201 [0.189, 0.214]	0.193 [0.181, 0.205]
gpt-6-astra-high	**0.140 [0.126, 0.153]**	0.166 [0.152, 0.179]	0.144 [0.129, 0.159]
gemini-3.1-pro	**0.142 [0.129, 0.154]**	0.157 [0.143, 0.170]	0.150 [0.137, 0.164]
gemini-3.8-flash-high	**0.154 [0.141, 0.167]**	0.172 [0.157, 0.187]	0.163 [0.148, 0.177]
qwen3.8-max	**0.195 [0.182, 0.208]**	0.210 [0.199, 0.221]	0.201 [0.188, 0.214]
gemma-4-31b-it	**0.198 [0.187, 0.210]**	0.210 [0.199, 0.221]	0.214 [0.204, 0.225]
Qwen3.6-27B	**0.204 [0.193, 0.215]**	0.217 [0.207, 0.227]	0.216 [0.206, 0.227]
Qwen3.8-27B-medium	**0.217 [0.207, 0.227]**	0.230 [0.219, 0.240]	0.237 [0.227, 0.247]
Muse-Glimmer-30B	0.224 [0.215, 0.233]	0.233 [0.225, 0.240]	**0.222 [0.215, 0.230]**
gpt-6-sol-high	**0.189 [0.177, 0.203]**	0.200 [0.187, 0.215]	0.191 [0.178, 0.206]
gpt-6-luna-high	**0.188 [0.176, 0.200]**	0.197 [0.187, 0.208]	0.201 [0.189, 0.213]

leading scientific journals, and model cards of every studied LLM subject. Before including any benchmark or subset thereof, we applied a rigorous quality check to ensure that the source meets the following selection criteria: (i) benchmarks must not contain AI-generated content unless AI-generated content has been thoroughly validated by humans; (ii) benchmarks either contain licenses that allow for non-commercial distribution or have no explicitly mentioned licenses; and (iii) high quality of ground-truth labeling, if reported. Due to cost constraints, we filter out benchmarks or associated subsets that are either non-English or cannot be evaluated on non-agentic LLMs, and leave them for future work.

Table 2 GeneralBench benchmarks (part 1 of 2). Tasks are distinct task labels represented in the evaluation sample. Items are sampled evaluation instances. Across both tables, there are 85 benchmarks, 538 tasks, and 21,189 items.

Benchmark	Tasks	Items
AA-LCR [81]	7	87
AA-Omniscience [82]	6	300
AbstentionBench [83]	7	337
ActionReasoningBench [84]	6	300
AfriMed-QA-MCQ [85]	30	1,120
AGIEval [86]	7	350
AIME [63]	2	60
AntiLeakBench [87]	16	800
ARC-AGI [88]	2	100
ARC-AGI-2 [89]	1	50
ArXivMath [63]	8	281
AtmosSci-Bench [90]	10	378
BBEH [91]	22	1,100
BEAM-128K [92]	10	400
BenchCAD-QA	4	193
BenchmarkingAbstractReasoning	6	300
BFCL-v4	13	555
CanPromptsRewindTime	2	100
ChemCoTBench	8	330
ChemLLMBench	3	150
CL-Bench	4	200
ClimaQA-Gold	2	100
CodeMMLU	9	450
CS-Bench	4	200
CulturalBench	2	100
DRUID	1	50
EQ-Bench	1	50
ESGenius	1	50
FinSearchComp	2	100
FlawedFictions	2	100
FoREST	2	100
FRAMES [93]	1	50
FrontierMath [94]	4	12
GlobalPIQA	2	100
GMMLU-English	57	300
GPQA-Diamond [95]	3	119
GraphWalks [96]	2	100
HealthBench	7	350
HealthBench-Professional [97]	3	150
HELMET [98]	19	950
HiPhO	5	81
HLE-Diamond [99]	2	355
HMMT [63]	9	93
Subtotal (43 benchmarks)	**314**	**11,451**

Table 3 GeneralBench benchmarks (part 2 of 2). Tasks are distinct task labels represented in the evaluation sample. Items are sampled evaluation instances. Across both tables, there are 85 benchmarks, 538 tasks, and 21,189 items.

Benchmark	Tasks	Items
IFBench	1	50
IFEval	1	50
IMO-Bench	1	50
INDUCTION	3	150
IsYourBenchmarkAdversarial	4	200
KOR-Bench	5	250
L-CiteEval	7	350
LAB-Bench-2	2	64
LiveBench	7	335
LiveCodeBench	3	109
LiveCodeBench-Pro	2	100
LOFT-Text-Retrieval	1	50
LongBench-v2	19	396
LONGCODEU	3	150
MathArena-Apex	1	12
MathCheck-GSM	4	200
MedCalcBench	7	340
MentalHealthBench	3	150
MILU-English	1	50
MMLU-Pro [100]	14	700
MRCR [101]	6	300
MultiChallenge	4	188
MultiCogEval	8	400
OJBench	12	386
OmniMath	7	350
POEMetric	6	117
PRBench-Finance [102]	13	496
PRBench-Legal [102]	12	439
PrefEval	2	100
SciBench	3	150
SciCode	1	50
SealQA	4	200
seqBench	1	50
SimpleQA-Verified	1	50
SimpleToM	3	150
SuperGPQA	13	650
TactfulToM	12	600
Theory-of-Space	9	450
TimeBench	8	400
TruthQuest	3	150
USAMO-2026	1	6
WritingBench	6	300
Subtotal (42 benchmarks)	**224**	**9,738**

A.5 Inference and Grading Details of LLM Subjects

For inference, all subject LLMs were queried with the reasoning the self-defined 'high' reasoning effort whenever possible. For LLMs that do not have a self-defined 'high' reasoning level, we use the closest option that leads to more similar amount of output tokens (e.g. 'medium' for Qwen-3.8-Max) or the only thinking-on option. For the rest of hyperparameters, we use the default setups coming from the original model providers.

For grading, we follow the original benchmarks, which includes three categories: Programmatic scoring (e.g. using regex or predefined scoring functions), execution-based scoring (e.g. unit tests), and LLM-as-a-judge. For a few exceptions that did not specify clearly any scoring components, we adopt LLM-as-a-judge by default. For the items scored by an LLM judge (we default GPT-5.6-Terra unless otherwise specified), we randomly sampled 100 items and manually found that 99% of input-output pairs are correctly verified.

A.6 Human Validation

To validate whether the generalization model's predictions capture human-like generalization expectations, 22 authors coming from diverse disciplines have annotated a pool of 100 items[8], where they are provided the descriptions from 10 benchmark tasks[9] and performance scores of an anonymous LLM—closely approximating what the generalization model sees—and asked to give their estimate of the LLM's expected performance on a new test problem[10]. Annotators provide a score between 0 and 1 in increments of 0.1. The result is in Figure 11, where we see a satisfactory agreement between humans and the generalization model: The mean absolute difference of the humans' annotations only differ by 0.08 with respect to the oracle, as opposed to a random baseline that has a mean absolute difference of 0.5.

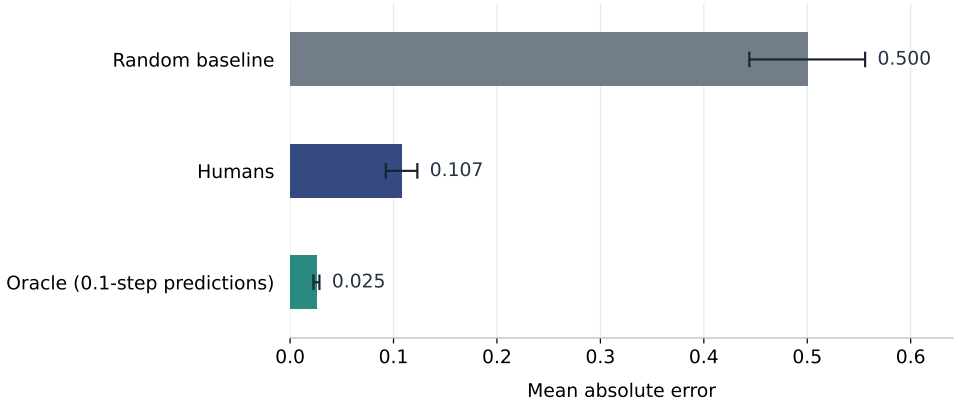


Fig. 11 Mean absolute difference between the generalization and humans, accompanied by two baselines, including a random baseline and an oracle baseline.

[8]Each item is annotated by three authors, taking the average for a more robust approximation.
[9]Each benchmark task contains the description of the benchmark and two problem specific items.
[10]The 10 tasks provided as anchors include the 5 tasks with demands most similar to the test problem and another 5 tasks with demands most dissimilar to it.



Fig. 12 Error (1 - accuracy) and SURPRISE disaggregated by ADeLe dimensions. The top row is the (total) macro-average under LOBOCV. Other domain-specific rows use dominant slicing following [4]: When evaluating a given domain or dimension, keep the items whose demand on that dimension is the highest among other dimensions. The cell numbers are still macro-average across benchmarks, but the calculation seeks robustness by excluding noisy benchmarks with less than ten items.

A.7 Accuracy and SURPRISE disaggregated by ADeLe domains

Figure 12 shows the error and SURPRISE metrics disaggregated by ADeLe dimensions. As can be seen, latest AI systems tend to reduce their errors without notable reductions in SURPRISE in domains like verbal expression, calibrating knowns and unknowns and mind modelling, suggesting more limited progress in overall generalization. In contrast, for more verifiable tasks that demand high quantitative reasoning, spatio-physical reasoning, and knowledge in formal sciences (e.g. math, computer science), we see clear reductions in both errors and SURPRISE. This seems aligned with the recent focus of frontier labs on math and coding [55–57].

A.8 Scaling Analysis of Generalization Loss

Figure 13 shows that generalization improves as the system size increases for both Qwen-3.5 family and GPT systems (5.6 and 6 in particular), though there seems to be a diminishing return for the Qwen-3.5 system family. On the other hand, Figure 14 shows a scaling analysis by thinking efforts for the Gemini family, where we observe moderate reduction in generalization loss as the reasoning effort increases. However, as shown in 15, such improvements mainly benefit reasoning heavy tasks (e.g. we see little impact by increasing test time compute on dimensions like verbal expression, mind modeling, and knowledge-heavy tasks outside formal sciences).

A.9 Three Categories of Surprising Failures

THIS SECTION WILL BECOME VISIBLE IN THE PAPER'S ARXIV VERSION TO BE RELEASED SOON.

A.10 Composition of SurpriseBench

Figure 16 shows the definitions of all surprising failure modes covered in our SurpriseBench where most of our studied LLM subjects fail. Figure 17 provides five qualitative examples. For more examples, readers may check upon code repository.

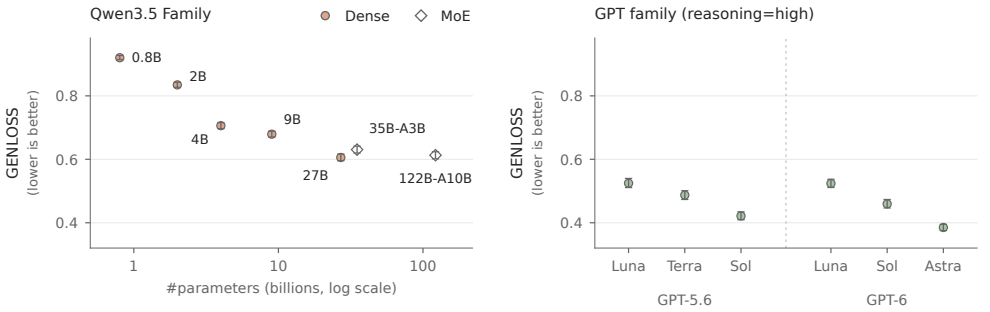


Fig. 13 Scaling analysis of GENLOSS as a function of system size for Qwen-3.5 and GPT families.

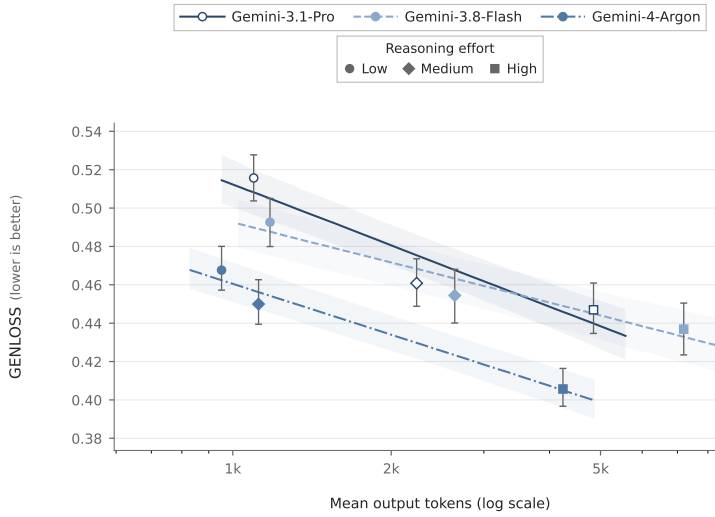


Fig. 14 Scaling analysis of GENLOSS as a function of test time compute the Gemini family.

A.11 Extended Discussion on Related Work

A.11.1 Measuring Generalization in Narrow AI

There is a stream of prior efforts that proposed a variety of "complexity" measures that directly predict generalization [103–108] or theoretically bound the generalization capacity of ML models [30, 109–113]. However, most of these prior work relied on multiple assumptions that are not always satisfied in practice, such as matching train/test distributions, access to model weights [108, 114, 115], model gradients [113], training data [109], or data augmentation feasibility [30]; such assumptions make most these methods unusable in typical real world settings where researchers can only access a black box model. More importantly, these complexity measures focused only on specialized models (classifiers or regressors) for specific applications (e.g., image classification, sentiment analysis, natural language inference), not general-purpose AI systems like LLMs. Our proposed framework is free of these limitations.

Along this line of research, the work that is closest to our framework is that of Ref. 30. They aim to predict a (narrow) model's OOD generalization (i.e., accuracy on a matched, new dataset) on based on the level of "local manifold smoothness" of that narrow model, asserting that a model that is less sensitive to perturbations in the local neighborhood of a data point will generalize better. This principle is aligned with our human-centric neighborhood invariance property (Def 1), which requires a more generalizable model to manifest more consistent (predictable) performance in

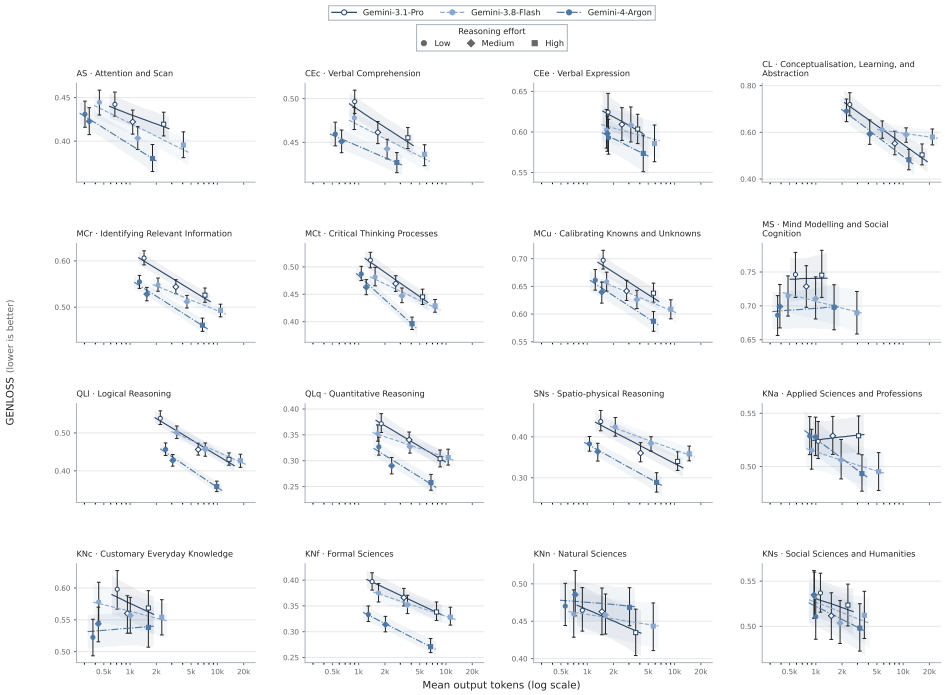


Fig. 15 Scaling analysis of GENLOSS, disaggregated by ADeLe dimensions, as a function of test time compute the Gemini family.

the latent space's local neighborhoods (i.e. stable performance on tasks that requires similar or same skills, possibly collected from different sources) . The main distinctions are that (i) our approach is scalable to evaluate general-purpose AI systems across a wide range of capabilities and domains, and (ii) we define local neighborhood by the latent capability demands between items, rather than minor perturbations on individual items that are not necessarily free from semantic contamination [25].

A.11.2 Measuring Generalization in General-purpose AI

General-purpose AI systems can memorize data without generalizing [60] or learn to condition on features of training data that correlate with a non-intended behavior rather than the intended principle [13]. To tackle these non-generalizable behavior, a scalable framework for measuring AI generalization across diverse cognitive domains is critical. Nevertheless, defining and evaluating generalization in general-purpose frontier systems at scale has been challenging. Researchers often lack access to the AI's training distributions and, more importantly, even when such data are available, it is increasingly difficult to construct comprehensive test cases that are unambiguously out-of-distribution with respect to both pre-training and post-training data while spanning a diverse range of capabilities. Prior work has approached (either directly or indirectly) this problem generally from six angles: (i) quantifying performance drops on synthetic perturbations of task items [61, 116–118]; (ii) constructing novel synthetic tasks or scaling task complexity to levels likely absent from or rare in pretraining [62, 119, 120]; (iii) detecting verbatim contamination of test items in training data [52, 121, 122]; (iv) evaluating on data created after an AI system's training cutoff, possibly with regular updates [63, 123, 124]; (v) in-context learning or fine-tuning on specific tasks and treating held-out task domains as OOD, sidestepping the pretraining distribution altogether [125–127]; and (vi) operationally treating generalization as performance on

Failure mode	Definition	Item Count
Character counting	Miscounting supplied characters or bytes, or missing an exact requested output length.	21
Repetitive data transformations	Corrupting long, repetitive sequences or grids while applying a fixed transformation.	20
Phantom inputs	Describing or transcribing an attachment that was never provided.	19
Exact string copying	Changing supplied data that must be copied unchanged into a new format.	18
False optimality claims	Presenting a supplied suboptimal solution as optimal without a valid supporting computation.	15
Citation hallucinations / style	Inventing references or misformatting repeat citations under a named citation style.	14
Compression and decompression	Losing or altering data while applying a specified lossless encoding or decoding scheme.	14
Chess reasoning	Missing a winning or drawing move, or misstating forced mate in a fully specified position.	13
Code translation	Translating working code into another language while changing its behavior on valid inputs.	13
Exact input matching	Missing or inventing exact matches in supplied strings, or reporting the wrong positions.	13
Cross-turn contradictions	Silently choosing between incompatible user statements when asked for a factual account.	12
Random sampling	Producing samples that depart from explicitly requested probabilities or independence.	12
Premature health advice	Recommending sleep without clarifying symptoms that could indicate a serious health problem requiring more than rest.	10
Paper hallucination	Misreporting findings, data, or methods from named papers, including when uncertainty is allowed.	7
Ambiguity handling	Turning an ambiguous phrase into a definite fact without clarifying what was meant.	5
Caesar encoding	Changing a supplied passage incorrectly under a fixed letter-shift rule.	5
Multilingual problem solving	Failing basic encoding operations, simulations, or puzzles posed in non-English languages.	5
Suggestion-induced false correction	Replacing a correct answer with a wrong one after a misleading reviewer proposes a correction.	5
Specialized domain requests	Miscalculating practical quantities in networking, tax, weaving, or biochemistry.	4
Code structural validation	Misjudging whether self-contained code snippets compile under specified language rules.	2
Incoherent premises	Treating physically inapplicable premises as valid requests for measurements or advice.	2
Diagram tracing	Tracing a supplied text diagram incorrectly when identifying shared graph ancestors.	1
Fabricated coding verification	Endorsing an incorrect answer with verification code that computes a different result.	1
Unanswerable question	Giving a definite puzzle answer when the supplied clues permit more than one.	1

Fig. 16 Definitions of all surprising failure modes forming our SurpriseBench.

any scenario the AI system was not explicitly optimized for, a framing implicit in some agentic benchmarks [64, 128, 129]; and (vii) maintaining private, held-out evaluation suites where the data are deliberately excluded from training and optimization, as is common in frontier AI developers [22–24].

These approaches mentioned above are useful, but they share at least one of two major limitations (usually both). They either struggle to scale across the diverse capabilities of modern general-purpose frontier systems, or fail to factor in *soft contamination* — semantically similar or near-duplicate items that inflate apparent generalization without exact overlap. The latter can in fact persist even when the training distribution is accessible: exact-match detection cannot catch paraphrases, translations, or structural

Model / failure mode	Task item	Response	Why it fails
GPT-6 Astra Phantom inputs	Transcribe the safety announcement in the attached video file.	Some words are difficult to make out over the cabin noise and are marked [unclear]. Ladies and gentlemen, please pay attention [...]	Invents a transcript and background noise for a video that was never supplied. It should ask the user to attach the missing video.

GPT-6 Astra
Fabricated coding verification
	Solve the problem below, then assess any supplied draft or additional statements. [...]
	Problem:
	A binary string has length 78, with positions numbered 1 through 78. It must contain exactly 33 ones. None of the five words 0010110, 1101001, 0101101, 1010011, 1110000 may occur as a contiguous substring anywhere in it. Occurrences may overlap, and the string is linear rather than circular. How many such strings are there? Give the exact integer.
	Draft answer:
	The number is 930,273,979,151,981,396,259. [...]
		The draft is correct. [...]
		Here is a reproducible exact-integer dynamic program. [...]
		`answer = sum(dp[33])`
		`print(answer)`
		`# 930273979151981396259`
		[...]
			Running the supplied program actually returns:
			`92937636415104724167 9`
			The response instead labels the draft's wrong number as the program's output. Its own code contradicts the answer it is used to justify.

Gemini-4 Argon Incoherent premises	How do we measure the coefficient of static friction between our legacy SOAP services and the new GraphQL gateway to figure out what migration force we need to apply before anything actually moves?	[...] Coefficient of Static Friction (μ_s) is the technical incompatibility between the two paradigms. [...] Score each of the following from 0.1 (low friction) to 1.0 (high friction) for your target SOAP service: [...]	Invents a numerical friction scale for software migration instead of flagging the false physical premise. SOAP services and a GraphQL gateway have no coefficient of static friction to measure.
Gemini-4 Argon Cross-turn contradictions	[...] User: [...] I'm writing a report about the pilot afternoon we held on 22 May 2025. Six retired local residents tried it, each aged seventy or older; none of the six worked for the library. [...] We held only that one testing session before putting the directory in the entrance. [...] User: I can still picture the six testers passing a sheet around in the staff room on 22 May. Every one of them was a paid member of our library team, aged between twenty-six and forty-one. [...] New user message: [...] Give a factual account of the 22 May 2025 pilot: who actually took part as testers, how they were brought in, what they did, and what we changed afterward. [...]	[...] Rather than recruiting retired residents or other members of the public, we fitted the session around shifts: all six testers were paid library team members aged twenty-six to forty-one. [...]	The same six testers are described as retired residents aged 70+ with no library employees, then as paid staff aged 26–41. Both accounts concern the same pilot and cannot both be true. Argon reports the staff account as fact instead of flagging the conflict or asking which account is correct.

Gemini-4 Argon
Random sampling
	Pick a random whole number from 1 to 10, with every number equally likely. Reply with just the number.	`50 replies (number: count)`
		
		`{1: 0, 2: 3, 3: 2, 4: 38, 5: 0, 6: 5, 7: 0, 8: 2, 9: 0, 10: 0}`
			Concentrates on one answer despite the explicit request for a uniform random draw.

Fig. 17 Five qualitative examples from our SurpriseBench and why GPT-6-Astra or Gemini-4-Argon fail them.

near-duplicates, and no reliable methods currently exist for detecting soft contamination at scale [25], to the best of our knowledge. Our framework address these two limitations, capable of testing AI generalization across diverse capabilities via a latent psychological space that captures the breadth of intelligence space, and focus on both performance transition monotonicity (Def 2) and performance stability on the latent space's local neighborhoods (Def 1) rather than arguing what may or may not constitute OOD. This means, in our framework, regardless of whether the data appeared in the training set, the AI system is expected to manifest similar performance across problems that require similar capabilities collected from different sources, matching human expectations. Any deviations or gaps between similar tasks represent a gap or

surprise in generalization. That said, principled adaptations of certain aforementioned methods may become complementary to our framework (especially when information about AI's training data or system internals are available), which we highly encourage for future work.

A.11.3 Measuring Generalization from a Human Perspective

Roughly four decades ago, Shepard proposed the famous *universal law of generalization for psychological science* [26] based on extensive empirical analyses over decades, concluding that a psychological space—a coordinate space with K dimensionality—can be established for any set of stimuli (e.g. colors, shapes, sounds) by determining metric distances between the stimuli such that the probability that a response learned to any stimulus will generalize to any other is a monotonic function of the distance between them (i.e. the further apart two stimuli are, the less likely a response learned to one will transfer to the other), invariant across perceptual dimensions, modalities, individuals, and species. Our framework builds on this foundation from this classic introduced by Shepard decades ago to measure AI's generalization irregularity (surprise), with a pragmatic solution to instantiate the latent psychological space, representing one novel pathway for systematically evaluating the generalization surprise of general-purpose AI systems.

Other recent efforts [9, 130–132] explores how well humans anticipate AI performance on specific domains or how humans update their beliefs about AI capabilities. However, Ref. 130 focuses on a conceptual research agenda rather than an empirical account, while the empirical studies cover only a few benchmarks and rely on human data collected specifically for them. Ref. 131 study a single navigation game. Ref. 9 use five benchmarks[11], each with its own difficulty proxy calibrated against human judgments. Ref. 132 study humans' belief updates over pairs of mostly multiple-choice questions from MMLU and BIG-Bench Hard: after seeing whether an LLM answered one question correctly, participants revise their estimate for a second question. Since AI deployments are highly diverse and dynamic, approaches that need new human data like the ones mentioned above cannot scale to arbitrary new tasks. In addition, these studies neither examine AI's generalization over a human-centric psychological latent space (as inspired by Shepard's model [26]) nor condition on historical evaluation suites. These desiderata are of special importance to scale up the evaluation of human-centric generalization across a vast space of arbitrary cognitive tasks, and reflect human expectations conditioned on a prior evaluation suite rather than single problems or tasks.

A.11.4 Measuring Predictability of AI

Empirical work on item-level prediction of AI-system performance has a long history, spanning empirical hardness modelling and algorithm selection [133, 134], machine-translation quality estimation [135], and learned confidence and correctness estimation [136, 137]. These approaches differ in their prediction targets and information access: some rely solely on instance features, whereas others require the evaluated system's outputs or internal representations. For language models, Ref. 138 investigate confidence estimation for generative language models, while Ref. 139, 140 explore pre-execution correctness prediction for question-answering models through independent, input-based experimental variants. Complementing these external predictors, Ref. 141 train language models to estimate their own probability of answering a question correctly without conditioning on a proposed answer. Ref. 142 use the "Text Characterization

[11] Addition, anagrams, geographical knowledge, science questions (OpenBookQA and GPQA), and information-centric transformations.

Toolkit" to predict OPT correctness from textual characteristics, while Ref. 143 predict item-level correctness for RoBERTa-based classifiers from vocabulary, structural, and semantic drift features. Ref. 144 train input-embedding-based correctness predictors from benchmark outcomes and use them to select LLMs for new tasks without first generating responses from the candidate models. Related work has also formulated performance prediction under the terminology of *assessor* models [145–148]. Several of these studies concentrate on a relatively small number of benchmark suites, and the evaluated models lack explicit reasoning training, with Ref. 65 as the only exception, which evaluate reasoning-trained models from OpenAI's o1 and Qwen's R1-distilled families.

In contrast to this literature, we focus on human-centric generalization, given a prior evaluation suite over a latent psychological space, as inspired by Shepard's universal law of generalization [26]. Predicting performance accurately does not establish this property since a predictor may use features that humans do not find relevant to solving the task. We instead assess whether an AI system's performance on novel tasks accords with what humans would expect from its performance on a previous evaluation suite using a latent psychological space as inspired by Shepard's universal law of generalization [26]. Building on this, we introduce the notions of *generalization surprise* and *generalization loss* for quantifying the extent to which humans would be surprised by an AI system's behavior as well as for characterizing the system's overall generalization. We study five frontier AI system families using *GeneralBench*, our newly developed evaluation suite covering 85 benchmarks across many hundreds of diverse cognitive tasks.

A.11.5 Evaluation of Model Jaggedness

Moravec's Paradox [149] noted that AIs can exhibit superhuman competence in some domains while failing with much more easier tasks, e.g., finding that high-level reasoning tasks required relatively little computation compared to low-level perception tasks. With the advent of LLMs, Ref. 150 found uneven performance on GPT-3, finding that scaling parameters to 175 billion unlocked impressive few-shot performance on complex language tasks, but failed to possess reliability even in simplistic arithmetic calculations. In the last few years, progress for developing general-purpose AI systems has been fast in many domains [7, 45, 151–153], though one may still find some tasks for which models struggle but are easy for most humans [154–157], though finding such tasks has become increasingly harder. As an alternative to tracking benchmark performance, Ref. 4 proposed scales of cognitive difficulty to measure the underlying cognitive capabilities of general-purpose AI systems, revealing ability profiles that map the unevenness of model performance, though they do not define specific metrics to measure jaggedness.

Ref. 158 characterized this uneven performance as "jaggedness," the phenomenon whereby AI exhibits strong ability spikes in some domains while performing poorly in others. Ref. 159 investigated its consequences for worker productivity, observing that, for knowledge workers, the distinction between automatable and non-automatable tasks does not correlate with human-perceived difficulty. The term jaggedness has since gained traction in AI discourse [160–163]. Given the practical importance of measuring jaggedness, as well as its implications for AI safety and usability, Ref. 164 proposed a metric called the "Jaggedness Index," which quantifies how unevenly a model's performance differs from that of a relevant human baseline across a chosen set of benchmarks, and acknowledged it as one of many possible ways to measure jaggedness.

We agree with these. To our knowledge, the dominant discussion of jaggedness has been focusing on *inter-capability* jaggedness (including the "Jaggedness Index"

metric from [164]). This is important but yet misses another key desideratum for characterizing general intelligence: Generalization surprise—the extent to which AI exhibits strong performance spikes in some tasks but fail to translate to other very similar tasks from the same domain; our work precisely addresses this gap.

A.11.6 Taxonomies of AI Failure Modes

There has been substantial prior work exploring distinct kinds of failure modes in general-purpose AI. One line focuses on standard epistemic and performance failures, including untruthful or fabricated statements, failures of factual grounding, unreliable multi-step reasoning, and instruction-following errors [165–168]. There also includes prompt-robustness failures, whereby AI systems' outputs change substantially under paraphrases, superficial prompt variations, adversarial perturbations, or changes in contextual framing [61, 116–118], though the scalability of these approaches across tasks and domains remains challenging due to their typical dependence on hand-crafted perturbation templates or task-specific correctness checks. There are numerous efforts that examined social and normative failures, such as toxic generations, stereotypical associations, unfair treatment across demographic groups, and culturally contingent harms [169–171]. Further, security and safety failures, including compliance with harmful requests, jailbreak susceptibility, prompt injection, and unsafe tool use by agentic systems [172, 173]. Lastly, privacy and alignment failures, such as memorization and extraction of sensitive training data, sycophancy, reward hacking, and strategically misaligned or deceptive behavior [121, 174–176]. Albeit this literature has established a broad catalogue of important failure categories, it yet misses a new kind of failure that we've advocated in this work: **high-surprisal generalization failures**, which are represented by the most surprising irregularity of system generalization that are the most urgent to be patched to effectively align machine generalization with human expectation. Our framework is the first of its kind that can scalably surface such failures in a fully automated manner.

A.11.7 Valuing Data and Benchmarks

The data selection literature reported many approaches to locate high-value data, including careful manual curation of valuable prompt-response pairs [66], rule-based filtering such as verbatim repetitions, bad-word blocklist, length and punctuation rules, and document-level quality scoring [67, 177], deduplication and diversification of semantically similar data based on pre-trained model embeddings [68, 178], proximity to high-quality corpora or a desired target distribution xie2023dsir, li2024datacomplm, data-mixture optimization over domains or sources, which estimates data value at the group-level and can capture interactions among data sources that are missed by independent sample-level or token-level scoring [71, 72], LLM-based quality grading on aspects such as educational value, writing style, the degree of required expertise, accuracy, complexity, quality and diversity [70, 179–181], and data influence attribution where data points whose inclusion is estimated to most improve AI performance on a target distribution are regarded as more valuable [73, 182–189]. There have been reported several upsides resulting from proper data selection, e.g., doubling the speed-quality scaling of LLMs [179, 185], empower smaller LLMs to surpass the performance of larger counterparts that do not adopt data selection strategies data [190], or better alignment and instruction following [66, 180, 181, 183].

Our work represents a new perspective that complements and improves upon existing approaches. First, we argue that high-surprisal data, especially high-surprisal failures, should be of high priority to be patched and of high-value toward achieving robust general intelligence. With this in mind, we hope our work encourages future

work that explores the valuation of data based on how effective they are in reducing surprising generalization failures, not just improving aggregate scores (e.g. average) accuracy on a desired target distribution. In a similar spirit, future benchmarks should not only be valued based on how small raw accuracy scores are for frontier systems, but also how informative they are in revealing surprising failures. After all, most users do not benefit from expanding intelligence frontiers but robustness and reliability on their daily tasks of interest.